Machine Learning Based Smart Helmet for Personalized Construction Safety Monitoring

R26-SE-042

Project Proposal Report

Kariyapperuma M. V. – IT21200892

B.Sc. (Hons) Degree in Information Technology Specialized in Software Engineering

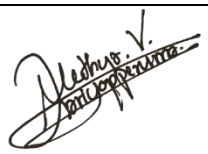
Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Kariyapperuma M. V.	**IT21200892**	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Dr. Kalpani Manuthunga		

ABSTRACT

The construction industry remains one of the most hazardous sectors globally, with workers exposed to risks such as extreme environmental conditions, toxic gases, excessive noise, and physical fatigue. Traditional safety approaches, including manual inspections and rule-based hazard identification methods, are largely reactive and fail to address dynamic and real-time safety challenges effectively. Although recent advancements in wearable sensor technologies have enabled continuous monitoring of worker conditions, most existing systems rely on fixed threshold-based alerts and lack intelligent, personalized safety analysis.

A key research gap identified is the absence of an integrated system that combines environmental hazard detection and worker physiological monitoring with adaptive, machine learning–based decision-making. Existing solutions often analyze these factors independently and do not account for individual worker differences, resulting in limited accuracy and reliability.

To address this gap, this research proposes an intelligent smart helmet system integrated with dual machine learning models for environmental hazard detection and personalized worker safety monitoring. The system collects real-time sensor data, processes it through a backend server, and applies machine learning techniques to generate adaptive safety risk predictions. A web-based dashboard is used to visualize data and provide real-time alerts to supervisors.

The proposed system utilizes technologies such as ESP32-based embedded systems, environmental and physiological sensors, Python-based machine learning frameworks, and web development tools. The expected outcome is a reliable, real-time safety monitoring system capable of improving hazard detection accuracy and reducing workplace accidents.

This research contributes to enhancing occupational safety, supporting Sustainable Development Goals related to health and safe working environments, and offering commercial potential for deployment in construction and industrial sectors.

Keywords: Smart Helmet, Construction Safety, Machine Learning, Wearable Sensors, Hazard Detection

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisor and academic staff for their continuous guidance, valuable feedback, and support throughout this research project. Their insights and encouragement have been instrumental in shaping the direction and quality of this work.

I am also thankful to the faculty members of the Software Engineering program for providing the necessary knowledge and academic foundation required to undertake this research. Their teaching has greatly contributed to the development of both technical and analytical skills.

In addition, I would like to acknowledge the support of industry professionals and peers who provided useful insights into construction safety practices and real-world challenges, which helped in refining the problem and proposed solution.

Finally, I extend my gratitude to my family and friends for their encouragement and support throughout the project.

TABLE OF CONTENTS

DECLARATION ii

ABSTRACT iii

ACKNOWLEDGEMENT iv

LIST OF TABLES vii

LIST OF FIGURES viii

LIST OF ABBREVIATIONS ix

LIST OF APPENDICES x

1. INTRODUCTION 1

1.1 Background and Context 1

1.2 Literature Review 2

1.3 Research Gap 4

2. OBJECTIVE 8

2.1 Main Objective 8

2.2 Specific Objectives 8

3. METHODOLOGY 9

3.1 Research Approach 9

3.2 Technical Methods and Algorithms 9

3.3 Tools and Platforms 10

3.4 Data Collection and Ethical Considerations 10

3.5 Validation Strategy and Performance Metrics 11

4. HIGH-LEVEL SYSTEM ARCHITECTURE 12

5. USER REQUIREMENTS 14

5.1 Stakeholder Identification 14

5.2 Functional Requirements 14

5.3 Non-Functional Requirements 14

6. COMMERCIALIZATION PLAN 16

6.1 Target Market and Customer Persona 16

6.2 Value Proposition 16

6.3 Revenue Model 16

6.4 Cost Estimation 17

6.5 Pricing Strategy 17
6.6 Competitive Advantage 17
6.7 Intellectual Property Considerations 17
7. BUDGET AND JUSTIFICATION 18
7.1 Hardware costs 18
7.2 Software licenses 18
7.3 Cloud services 19
7.4 Human effort costs 19
8. WORK BREAKDOWN STRUCTURE (WBS) 20
9. GANTT CHART 21
10. REFERENCES LIST 22
10. APPENDICES 26
10.1 Appendix I: Helmet Design 26

LIST OF TABLES

Table 1: Comparative Analysis of Existing Researches 5
Table 2: Hardware Components of Proposed Helmet 10
Table 3: Performance Metrics 11
Table 4: Hardware Costs 18

LIST OF FIGURES

Figure 2: High-Level System Diagram.....12

LIST OF ABBREVIATIONS

Application Programming Interface	-	API
Artificial Intelligence	-	AI
Bluetooth Low Energy	-	BLE
Design Science Research	-	DSR
Hypertext Transfer Protocol Secure	-	HTTPS
Long Range	-	LoRa
Machine Learning	-	ML
Internet of Things	-	IoT
Job Safety Analysis	-	JSA
Software as a Service	-	SaaS
Structured Query Language	-	SQL
Sustainable Development Goal	-	SDG
Ultraviolet	-	UV
Wearable Sensor Device	-	WSD

LIST OF APPENDICES

Appendix I: Helmet Design 126

1. INTRODUCTION

1.1 Background and Context

The construction industry plays a critical role in economic development; however, it remains one of the most hazardous industries globally, with high rates of accidents, injuries, and fatalities [1]. International Labour Statistics mentions that each year, around the world, around 60,000 fatal accidents occur on construction sites [2]. Accidents on construction sites are not only a threat to workers' lives but also cause expensive project delays, significant legal costs, and serious damage to the organization's reputation [3]. Construction hazards can be defined as any potential cause that poses a risk to worker safety or public health [4].

Traditional hazard identification approaches such as Job Safety Analysis (JSA), safety audits, and rule-based inspections are largely reactive, relying heavily on human judgment and predefined hazard lists [5]. These methods are time-consuming, fail to capture dynamic and unexpected risks, and often detect hazards only after incidents occur [6, 7, 8, 5]. This limitation has driven a shift toward proactive, data-driven safety approaches, including remote monitoring technologies. In Assessing and Mitigating Workplace Hazards in Construction, Adeyemo (2024) reports that shifting to a proactive, risk-based hazard identification system reduces workplace accidents by 40%, near-miss incidents by 35%, and compliance violations by 67% [9].

Among remote monitoring technologies for hazard identification, Wearable Sensor Devices (WSDs) have emerged as a promising solution due to their ability to provide real-time, individualized monitoring of environmental and physiological conditions [10, 11, 12]. WSDs are available in multiple forms such as helmets, wristbands, vests, and head-mounted devices, enabling continuous tracking of worker exposure to hazards. Despite their potential, these technologies are still in an early stage of adoption, particularly in real-world construction environments [13].

A key limitation identified in existing research is that most WSDs operate using fixed threshold-based alert mechanisms. While thresholds provide a basic level of safety monitoring, they do not account for the complex and dynamic nature of construction environments or individual worker differences. This often results in false alarms, missed early warnings, and reduced reliability, limiting user trust and system effectiveness [10].

Furthermore, existing studies tend to analyze environmental hazards and worker physiological conditions independently, without considering their combined effect on safety [14]. This fragmented approach restricts the ability to capture real-world risk scenarios where multiple factors interact simultaneously.

To address these challenges, this research proposes an intelligent smart helmet system integrated with machine learning models. The system combines environmental hazard detection and personalized worker safety monitoring to provide adaptive and real-time safety alerts. This research aligns with Sustainable Development Goal 3 (Good Health and Well-being) and SDG 8 (Decent Work and Economic Growth) by promoting safer working conditions.

1.2 Literature Review

Recent research in construction and industrial safety has increasingly focused on the adoption of WSDs and intelligent monitoring systems to enhance worker safety. These systems aim to transition from traditional reactive approaches toward proactive, real-time hazard detection and prevention.

A significant number of studies have explored IoT-enabled smart helmet systems. For example, an IoT-based smart helmet integrates environmental sensors such as gas, temperature, and light sensors with wireless communication modules to provide real-time hazard alerts and remote monitoring capabilities [15]. Similarly, hybrid communication-based smart helmet systems using BLE Mesh and LoRa technologies have been proposed to improve scalability, real-time communication, and worker location tracking in large industrial environments [16]. These systems demonstrate how communication technologies can enhance safety monitoring beyond basic sensing.

Several studies have extended WSDs by integrating multiple devices and multimodal sensing. For instance, integrated smart helmet and jacket systems combine environmental and physiological sensors to monitor gas exposure, body temperature, and worker health conditions, providing comprehensive safety monitoring in hazardous environments [17]. Similarly, mining-focused wearable systems incorporate gas sensors, pulse sensors, and oxygen monitoring to improve worker safety in underground environments [18]. These approaches highlight the importance of combining multiple sensor types for more accurate hazard detection.

In addition to helmet-based systems, broader wearable sensing research emphasizes the role of personalized and continuous monitoring. WSDs can collect both physiological data (heart rate, activity levels) and environmental exposure data (air quality, temperature, noise), enabling the development of personalized safety and health models [19]. Long-term monitoring studies further demonstrate that combining environmental and physiological data can improve understanding of human comfort and risk perception in real-world environments [20].

Recent advancements have also introduced machine learning and AI-based approaches for intelligent safety monitoring. For example, SafeHelmet-AI utilizes multimodal sensor data and a CatBoost machine learning model to detect abnormal conditions such as heat stress, gas exposure, and falls in real time [21]. Similarly, embedded systems using TinyML techniques enable on-device activity recognition and fall detection, reducing latency and enabling real-time decision-making without reliance on cloud processing [22]. In addition, AI-based helmet systems using deep learning have been developed to detect safety compliance, monitor worker health, and identify hazards such as fire and abnormal physiological conditions [23].

Despite these advancements, several limitations remain in existing research and systems. First, many WSD solutions still rely on threshold-based alert mechanisms, where alerts are triggered only when predefined limits are exceeded. This approach fails to capture the dynamic and complex nature of construction environments and often results in false alarms or missed early warnings. Second, although some systems integrate multiple sensors, most studies still analyze environmental hazards and physiological conditions separately, limiting their ability to capture real-world risk scenarios where multiple factors interact simultaneously. Third, while machine learning has been introduced, its application is often limited to specific tasks such as anomaly detection or activity recognition, rather than providing a fully integrated and adaptive safety monitoring system.

Furthermore, many proposed systems remain at the prototype or experimental stage, with limited real-world deployment and validation. Challenges such as scalability, long-term reliability, user acceptance, and integration with existing safety management systems are not adequately addressed in current research.

Overall, the literature indicates significant progress in WSDs and intelligent monitoring systems. However, there remains a clear need for integrated, adaptive, and personalized safety monitoring solutions that combine environmental and physiological data using advanced machine learning techniques for real-time decision-making in construction environments.

1.3 Research Gap

Although significant progress has been made in WSDs, existing research and systems still exhibit several critical limitations that restrict their effectiveness in real-world construction environments.

Comparative Analysis of Existing Systems

The following table summarizes key features of selected systems identified in the literature:

Table 1: Comparative Analysis of Existing Researches

System / Study	Environmental Monitoring	Physiological Monitoring	Machine Learning	Real-time Alerts	Personalization	Integrated Risk Analysis
IoT Smart Helmet (2025) [15]	✓	✕	✕	✓	✕	✕
BLE-LoRa Smart Helmet [16]	✓	✕	✕	✓	✕	✕
Smart Helmet and Jacket System [17]	✓	✓	✕	✓	✕	✕
Coal Miner Wearable Device [18]	✓	✓	✕	✓	✕	✕
SafeHelmet-AI [21]	✓	✓	✓	✓	✕	Partial
AI Deep Learning Helmet [23]	✓	✓	✓	✓	✕	✕
Smart Wear Guardian [22]	✓	✓	✓	✓	✕	✕
Personalized Wearable Sensors Study [19]	✓	✓	✓	✕	✓	✕
Thermal Comfort Wearable Study [20]	✓	✓	✓	✕	✓	✕
Proposed System	✓	✓	✓	✓	✓	✓

Identified Research Gaps

1. Lack of Integrated Risk Analysis: Most existing systems focus on either environmental monitoring or physiological monitoring independently. Even in systems where both types of data are collected, they are not effectively combined to produce a unified safety assessment. As a result, these systems fail to represent real-world scenarios where multiple risk factors interact simultaneously.

2. Limited Personalization: Current systems largely rely on generic threshold values for all workers. However, physiological responses such as heart rate and body temperature vary significantly between individuals. The absence of personalized safety models reduces accuracy and may lead to false alarms or missed critical conditions.

3. Over-reliance on Threshold-Based Alerts: Many IoT-based safety systems operate using predefined thresholds. While simple to implement, this approach cannot capture complex patterns or predict risks in dynamic environments. Studies show that threshold-based systems often generate unreliable alerts and lack predictive capabilities.

4. Partial Use of Machine Learning: Although some recent systems incorporate machine learning, its application is often limited to specific tasks such as anomaly detection or activity recognition. There is a lack of comprehensive frameworks that utilize machine learning to perform holistic safety analysis combining multiple data sources.

5. Lack of Fully Integrated Intelligent Systems: Existing research tends to address safety problems in a fragmented manner, focusing on isolated features such as fall detection, gas detection, or compliance monitoring. Few systems provide a complete end-to-end solution that integrates sensing, intelligent analysis, and decision-making into a unified architecture.

The limitations identified above indicate that current approaches are not capable of addressing the complex, dynamic, and multi-dimensional nature of construction site hazards. Real-world safety risks arise from the interaction of environmental conditions and worker physiological states, which cannot be effectively managed using isolated monitoring or static thresholds. Furthermore, the lack of personalization reduces system reliability, while limited use of machine learning restricts predictive capabilities.

Proposed Research Contribution and Novelty

This research addresses the identified gaps by proposing:

- A dual-model machine learning framework
- Integration of environmental hazard detection and personalized worker safety monitoring
- Use of worker-specific baseline data for adaptive alerts
- A risk fusion layer to generate a unified safety assessment

The key novelty lies in combining:

1. Multi-source data integration (environmental + physiological)
2. Adaptive, personalized safety monitoring
3. Intelligent decision-making using machine learning

Existing literature demonstrates that WSDs significantly enhance construction safety through real-time monitoring of environmental and physiological conditions, while machine learning techniques improve the ability to predict potential risks. In addition, personalized models have been shown to increase the accuracy of safety assessments by accounting for individual differences in worker health and responses. However, current research lacks a unified system that effectively integrates real-time monitoring, machine learning–based prediction, and personalized safety analysis within a single framework. Therefore, this study addresses a critical gap by proposing an integrated intelligent safety system that aligns with modern construction safety requirements and Industry 4.0 advancements

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate a machine learning–based smart helmet system that integrates environmental hazard detection and personalized worker safety monitoring to generate real-time safety risk predictions with at least 85% accuracy under simulated construction conditions.

2.2 Specific Objectives

1. To design and implement a smart helmet prototype equipped with environmental and physiological sensors capable of collecting real-time data.
2. To develop and train two machine learning models for environmental hazard detection and personalized worker safety analysis.
3. To design and implement a backend system and web-based dashboard for real-time data processing, visualization, and automated safety alert generation.
4. To evaluate the performance of the proposed system in terms of prediction accuracy, response time, and reliability using experimental testing scenarios.

3. METHODOLOGY

3.1 Research Approach

This research follows a Design Science Research (DSR) approach combined with an experimental methodology. The Design Science approach is used to design and develop an innovative artifact, namely the intelligent smart helmet system. The experimental approach is applied to evaluate the performance of the proposed system under controlled conditions. This combination is appropriate as the study involves both system development and empirical validation.

3.2 Technical Methods and Algorithms

Data Processing

Sensor data collected from the helmet are preprocessed through:

- Data cleaning (removal of noise and missing values)
- Normalization of sensor values
- Feature engineering (e.g., heart rate deviation from baseline)

Machine Learning Models

Two models are developed:

1. **Environmental Hazard Detection Model**
    - Inputs: Gas levels, ambient temperature, noise level, UV exposure
    - Algorithms: Random Forest or Gradient Boosting
    - Output: Environmental risk level (Safe, Warning, Critical)
2. **Personalized Worker Safety Model**
    - Inputs: Heart rate, body temperature, baseline deviations, work duration
    - Algorithms: Random Forest or Support Vector Machine
    - Output: Worker health risk level

Risk Fusion Mechanism

Outputs from both models are combined using a rule-based or weighted scoring approach to generate a final safety risk level.

3.3 Tools and Platforms

Hardware

Table 2: Hardware Components of Proposed Helmet

Ambient Temperature	:	DHT11
UV exposure	:	GUVA-S12SD UV detection sensor module
Gas/ VOC	:	MQ2 Gas sensor
Noise	:	INMP441 MEMS microphone
Body Temperature and	:	LM35
Pulse Rate	:	MAX30102
Location	:	Neo-6M GPS module
Processing and Control Unit	:	Node MCU ESP32 microcontroller
Battery	:	Lithium-ion battery unit

Software and Development Tools

- Python (for machine learning model development)
- Scikit-learn / TensorFlow (for model training and evaluation)
- Node.js (backend API development)
- MongoDB or PostgreSQL (data storage)
- React.js (web dashboard development)

3.4 Data Collection and Ethical Considerations

Data Collection

Data will be collected through the smart helmet prototype in controlled experimental scenarios. Participants will wear the helmet while performing simulated construction activities. The dataset will include:

- Environmental data (temperature, gas levels, noise, UV)
- Physiological data (heart rate, body temperature)
- Contextual data (worker ID, timestamp, activity duration)

Ethical Considerations

Since the research involves human participants and physiological data:

- Informed consent will be obtained before data collection

- Personal data will be anonymized
- Data will be used strictly for research purposes
- Ethical clearance will be obtained from the relevant institutional committee if required

3.5 Validation Strategy and Performance Metrics

Validation Strategy

The system will be evaluated through experimental testing using collected datasets. The performance of the machine learning models will be validated using techniques such as:

- Train-test split
- Cross-validation

Performance Metrics

The following metrics will be used:

Table 3: Performance Metrics

For Machine Learning Models	For System Performance
• Accuracy • Precision and Recall • F1-score	• Response time for alerts • Reliability of data transmission • System usability and effectiveness

The overall system will be considered successful if it achieves high prediction accuracy and provides timely and reliable safety alerts.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

This research proposes an intelligent smart safety helmet system integrated with machine learning models and a web-based monitoring platform to improve construction worker safety. The system collects environmental and physiological sensor data from workers and analyzes it using machine learning to generate adaptive and personalized safety alerts.

Unlike existing systems that rely on simple threshold-based alerts, the proposed solution integrates two machine learning models that analyze environmental hazards and worker health conditions together. This allows the system to detect safety risks more accurately and provide early warnings to supervisors and workers.

The system consists of five main components:

1. Smart Helmet (data acquisition layer)
2. Communication Module (data transmission)
3. Backend Server (data processing and storage)
4. Machine Learning Models (intelligent analysis)
5. Web Application (visualization and alerts)

These components work together to collect, process, analyze, and present safety-related information in real time.

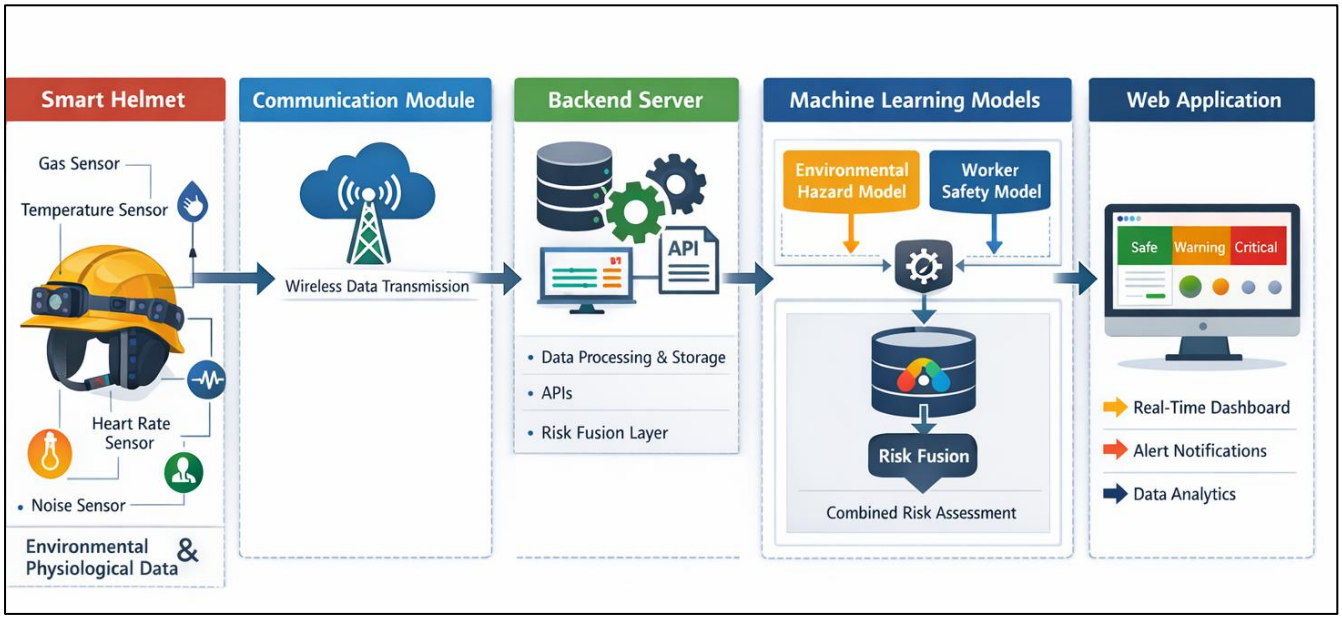


Figure 1: High-Level System Diagram

1. Smart Helmet (Data Acquisition Layer): The smart helmet serves as the primary data acquisition component of the system. It is equipped with multiple environmental and physiological sensors that continuously monitor both the surrounding conditions and the worker's health status. Environmental sensors collect data such as gas concentration, ambient temperature, noise levels, and UV exposure, while physiological sensors measure heart rate and body temperature. These sensors are connected to a microcontroller, such as the ESP32, which is responsible for aggregating, formatting, and transmitting the collected data. The helmet operates in real time, capturing data at regular intervals and preparing structured data packets that include sensor readings, timestamps, and worker identifiers.

2. Communication Module (Data Transmission Layer): The communication module is responsible for transmitting the collected sensor data from the smart helmet to the backend server. This is typically achieved through wireless communication technologies such as WiFi or mobile networks.

3. Backend Server (Processing and Storage Layer): The backend server acts as the central processing unit of the system. It receives incoming data from multiple helmets and performs several key functions, including data preprocessing, storage, and coordination of machine learning analysis. Upon receiving data, the backend performs cleaning and normalization to remove noise and ensure consistency. It also generates additional features, such as deviations from baseline physiological values. The processed data is then stored in a database for both real-time access and historical analysis.

4. Machine Learning Layer (Intelligent Analysis Layer): The machine learning layer is responsible for analyzing the processed data and generating safety predictions. It consists of two main models operating in parallel. The environmental hazard detection model evaluates environmental sensor data to determine whether the surrounding conditions are safe or hazardous. At the same time, the personalized worker safety model analyzes physiological data in relation to individual baseline values to detect abnormal health conditions such as fatigue or stress. The outputs of these models are combined in the risk fusion layer, which integrates both environmental and physiological risk factors to produce a final safety risk level.

5. Web Application (Presentation and Alert Layer): The web application serves as the user interface of the system, providing real-time visualization and monitoring capabilities. It displays sensor data, risk levels, and alerts in an intuitive dashboard format.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The system involves several key stakeholders:

- Construction Workers: Primary users who wear the smart helmet. They benefit from real-time safety monitoring and alerts.
- Site Supervisors / Safety Officers: Monitor worker conditions through the dashboard and take necessary safety actions.
- Construction Companies / Management: Use the system to improve safety compliance, reduce accidents, and enhance productivity.
- System Administrator: Responsible for managing system operations, user access, and data handling.

5.2 Functional Requirements

1. The system shall collect real-time environmental and physiological data from the smart helmet sensors.
2. The system shall transmit sensor data from the helmet to the backend server.
3. The system shall preprocess and store collected data in a database.
4. The system shall analyze data using machine learning models to detect safety risks.
5. The system shall generate a final risk level by combining model outputs.
6. The system shall display real-time data and risk levels on a web dashboard.
7. The system shall generate alerts when unsafe conditions are detected.
8. The system shall maintain historical records of sensor data and safety events.
9. The system shall allow supervisors to monitor multiple workers simultaneously.
10. The system shall support user authentication and access control.

5.3 Non-Functional Requirements

Security

- The system shall ensure secure data transmission using encryption protocols (e.g., HTTPS).
- The system shall protect user data through authentication and authorization mechanisms.
- Sensitive data shall be anonymized where necessary.

Performance

- The system shall process and display data with minimal latency.
- The system shall handle multiple concurrent users and devices efficiently.

Usability

- The dashboard shall provide a simple and intuitive user interface.
- Alerts shall be clear and easy to understand for quick decision-making.

Reliability

- The system shall ensure consistent data transmission and minimal downtime.
- The system shall handle errors and data loss gracefully.

Scalability

- The system shall support the addition of multiple helmets and users without performance degradation.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

Target Market

- Construction companies and contractors
- Infrastructure development firms
- Industrial sectors such as mining and manufacturing
- Occupational health and safety organizations

Customer Persona

- **Site Safety Manager / Supervisor:** Responsible for ensuring worker safety and compliance. Needs real-time monitoring and reliable alerts to prevent accidents.
- **Construction Company Management:** Focused on reducing accident-related costs, improving productivity, and meeting regulatory requirements.

6.2 Value Proposition

The proposed system provides a smart, data-driven safety solution that goes beyond traditional monitoring methods. It offers:

- Real-time safety monitoring using wearable sensors
- Intelligent risk prediction using machine learning
- Personalized safety alerts based on worker conditions
- Reduced accidents and improved workplace safety

This enables organizations to prevent incidents rather than react to them, improving both safety and operational efficiency.

6.3 Revenue Model

- **Hardware Sales Model:** One-time sale of smart helmet devices to organizations
- **Subscription Model (SaaS):** Monthly or yearly subscription for access to the web dashboard, analytics, and alerts
- **Maintenance and Support Services:** Optional service fees for system maintenance and updates

6.4 Cost Estimation

Initial Development Costs

- Hardware components (sensors, microcontroller, helmet integration)
- Software development (backend, ML models, web application)
- Testing and deployment

Operational Costs

- Cloud hosting and database services
- Maintenance and updates
- Customer support

6.5 Pricing Strategy

- Competitive pricing based on cost-plus strategy
- Tiered pricing for different company sizes (small, medium, large)
- Bundle pricing (helmet + software subscription)

6.6 Competitive Advantage

The proposed system offers several advantages over existing solutions:

- Integration of environmental and physiological monitoring
- Use of machine learning for predictive safety analysis
- Personalized alerts based on worker baseline data
- Real-time decision support through a centralized dashboard

These features provide a more accurate, adaptive, and comprehensive safety solution compared to traditional and existing systems.

6.7 Intellectual Property Considerations

- The system design, software architecture, and machine learning models can be protected through intellectual property rights
- Potential for patent protection on the integrated dual-model safety system
- Software components can be protected under copyright laws
- Data privacy and compliance with regulations must be ensured

7. BUDGET AND JUSTIFICATION

7.1 Hardware costs

Table 4: Hardware Costs

Component	Estimated Price (LKR)
MQ2 Gas sensor	395
DHT 11 ambient temperature	180
GUVA-S12SD light level sensor	1,200
KY 037 sound module	240
MAX30102 HRV sensor	790
LM35 body temperature sensor	165
NEO 6M GPS sensor	1,290
Emergency push button	190
LED RGB module	90
ESP 32 wroom	1,390
battery	950
Power button	110
Other	5,000
Total price	**11,900**

7.2 Software licenses

The project will primarily use open-source technologies, therefore no major licensing costs are expected.

- Python (Free)
- Scikit-learn / TensorFlow (Free)
- Node.js (Free)
- MongoDB / PostgreSQL (Free versions available)

Estimated Cost: 0 LKR

7.3 Cloud services

Cloud infrastructure is necessary for real-time data processing, storage, and remote monitoring through the web application. It ensures scalability and accessibility of the system.

- Cloud hosting (e.g., AWS / Firebase / Vercel)
- Database hosting
- API services

Estimated cost for development phase: **Approx. 2,000 – 5,000 LKR**

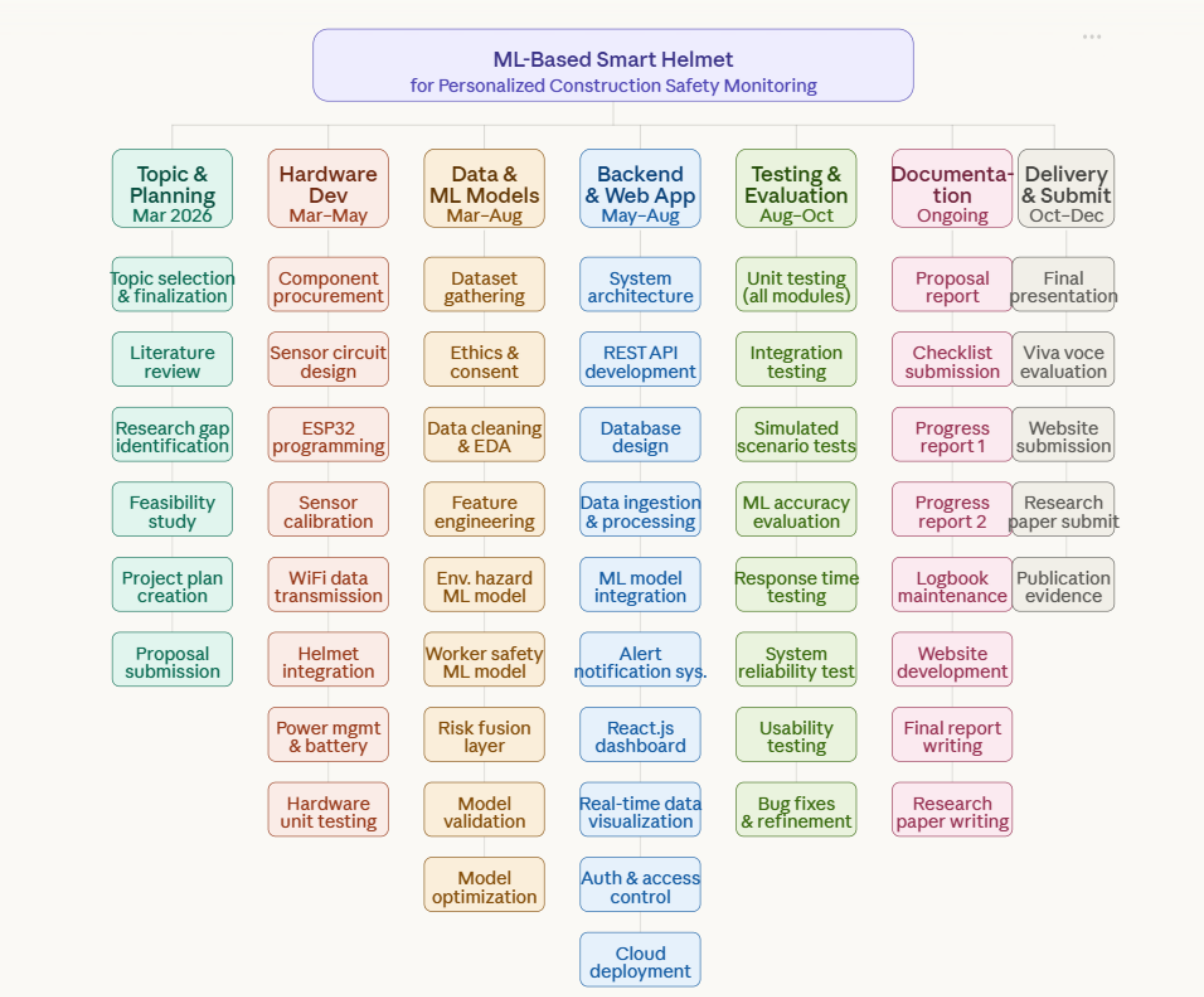
7.4 Human effort costs

The development of the system requires significant time and technical expertise across multiple domains, including embedded systems, machine learning, and web development. However, the estimated effort includes:

- System design and development
- Machine learning model development
- Hardware integration and testing
- Documentation and evaluation

Estimated effort: **300+ hours of development work**

8. WORK BREAKDOWN STRUCTURE (WBS)



9. GANTT CHART



10. REFERENCES LIST

[1] P. Forsythe, "Proactive construction safety systems and the human factor," in *Institution of Civil Engineers*, 2014.

[2] International Labour Statistics, "Occupational Safety and Health Management in the Construction Sector," 2021. [Online]. Available: https://www.ilo.org/meetings-and-events/occupational-safety-and-health-management-construction-sector-3.

[3] M. T. Newaz, M. Ershadi and M. Jefferies, "A critical analysis of construction incident trends and strategic interventions for enancing safety," in *Safety Science*, 2025.

[4] V. H. P. Vitharana, G. H. M. J. S. De Silva and S. De Silva, "Health hazards, risk and safety practices in construction sites – a review study," in *ENGINEER*, 2015.

[5] H. Kim, H.-S. Lee, M. Park, B. Chung and S. Hwang, "Automated hazardous area identification using laborers' actual and optimal routes," in *Automation in Construction*, 2016.

[6] G. Carter and S. Smith, "Safety Hazard Identification on Construction Projects," in *Journal of Construction Engineering and Management*, 2006.

[7] Y. M. Goh and . D. K. H. Chua, "Case-Based Reasoning Approach to Construction Safety Hazard Identification: Adaptation and Utilization," in *Journal of Construction Engineering and Management*, 2009.

[8] H. S. Kim, H.-S. M. Lee, M. M. Park, B. Chung and S. S. Hwang, "Information Retrieval Framework for Hazard Identification in Construction," in *Journal of Computing in Civil Engineering*, 2013.

[9] J. O. Adeyemo, " Assessing and Mitigating Workplace Hazards in Construction: A Risk Based Approach," in *International Journal of Research Publication and Reviews*, 2024.

[10] I. Awolusi, E. Marks and M. Hallowell, "Wearable technology for personalized construction safety monitoring and trending: Review of applicable devices," in *Automation in Construction*, 2018.

[11] H. Chen, Y. Mao, Y. Xu and R. Wang, "The Impact of Wearable Devices on the Construction Safety of Building Workers: A Systematic Review," in *Sustainability*, 2023.

[12] H. Fugate and H. Alzraiee, "Quantitative analysis of construction labor acceptance of wearable sensing devices to enhance workers' safety," in *Results in Engineering*, 2023.

[13] Y. Zhang, "Wearable Sensors: An ACS Sensors Special Issue," *ANC Sensors,* 2025.

[14] I. Awolusi, C. Nnaji, E. Marks and M. Hallowell, "Enhancing Construction Safety Monitoring through the Application of Internet of Things and Wearable Sensing Devices: A Review," *ASCE International Conference on Computing in Civil Engineering 2019,* 2019.

[15] G. C. Wong, N. Anizam, N. H. Nasruddin and C. Z. C. Hasan, "An IoT-Enabled Smart Safety Helmet for Enhancing Worker Protection on Construction Sites," *Politeknik & Kolej Komuniti Journal of Engineering and Technology,* 2025.

[16] G.-J. Kim, M. Cho, S.-J. Lim, T.-H. Kim, G.-H. Cho, Y.-S. Cho and S.-H. Park, "Design and Implementation of a Hybrid BLE Mesh–LoRa Smart Safety Helmet System for Port Logistics Environments," *Asia-pacific Journal of Convergent Research Interchange,* 2026.

[17] A. Raghuvanshi, P. Parikh and P. Shah, "Integrated Smart Helmet and Jacket System for Enhanced Worker Safety and Health Monitoring in Hazardous Environments," *International Journal Of Novel Research And Development,* 2025.

[18] J. K., S. K. S., S. B., S. S. V. and R. R., "Smart Wearable Device for Enhancing Safety and Efficiency of Coal Miners," *Journal of Innovative Image Processing,* 2024.

[19] O. Panahi, "Wearable Sensors and Personalized Sustainability: Monitoring Health and Environmental Exposures in Real-Time," *European Journal of Innovative Studies and Sustainability,* 2025.

[20] V. M. Gnecco, I. Pigliautile and A. L. Pisello, "Long-Term Thermal Comfort Monitoring via Wearable Sensing Techniques: Correlation between Environmental Metrics and Subjective Perception," *Sensors,* 2023.

[21] K. M. M., L. S., P. P. and S. M., "Smart Helmet–Based Wearable Intelligence for Interpreting Worker Health and Safety Conditions," *International Journal for Multidisciplinary Research,* vol. 8, no. 2, 2026.

[22] T. P, P. T and K. P, "Smart Wear Guardian," in *ITM Web of Conferences*, 2026.

[23] N. S, P. N. G, R. K and B. Steena, "Intelligent Insights into Safety Helmet Usage with Deep Learning," *International Journal of Advanced Multidisciplinary Research and Studies ,* 2025.

[24] A.Retalis, "Geographic information systems and science," *The Photogrammetric Record,* p. 396, 2005.

[25] N. E. E.-e. El.Faouzi, "Classifiers and distance-based evidential fusion for road travel time estimation," *Proceedings of SPIE, the International Society for Optical Engineering/Proceedings of SPIE,* 2006.

[26] Clunder, G A; Bass, A P; De Beek, F J;, "A long term travel time prediction algorithm using historical data," *PROCEEDINGS OF THE 14TH WORLD CONGRESS ON INTELLIGENT TRANSPORT SYSTEMS (ITS), HELD BEIJING, OCTOBER 2007,* 2007.

[27] W. W. H. "Machine Learning Methods in the Environmental Sciences: Neural Networks and," 2009.

[28] T. Bolu, "Real-Time Data Processing in AI Applications," *ResearchGate,* 2024.

[29] Nagy, Balázs;, "A New Method of Improving the Azimuth in Mountainous Terrain by Skyline Matching," *PFG – Journal of Photogrammetry Remote Sensing and Geoinformation Science,* vol. 88, no. 2, p. 121, 2020.

[30] Fedorov, Roman; Martinenghi, Davide; Tagliasacchi, Marco; Castelletti, Andrea;, "Exploiting User Generated Content for Mountain Peak Detection," *Electronic workshops in computing,* 2013.

[31] La Salandra, Antonio; Fraternali, Piero; Frajberg, Darian;, "A Location-Based Virtual Reality Application for Mountain Peak Detection," 2018.

[32] Cam, C;, "Navigation | Map, Compass and GPS | The Hiking Life," 11 October 2021. [Online]. Available: https://www.thehikinglife.com/hiking-and-backpacking-skills/navigation/.

[33] Macdonald, Makai;, "How Weather Influences Trail Conditions for Hikers," 07 May 2024. [Online]. Available: https://www.macfishes.com/impact-of-weather-on-trail-conditions/. [Accessed 24 August 2024].

10.APPENDICES

10.1 Appendix I: Helmet Design

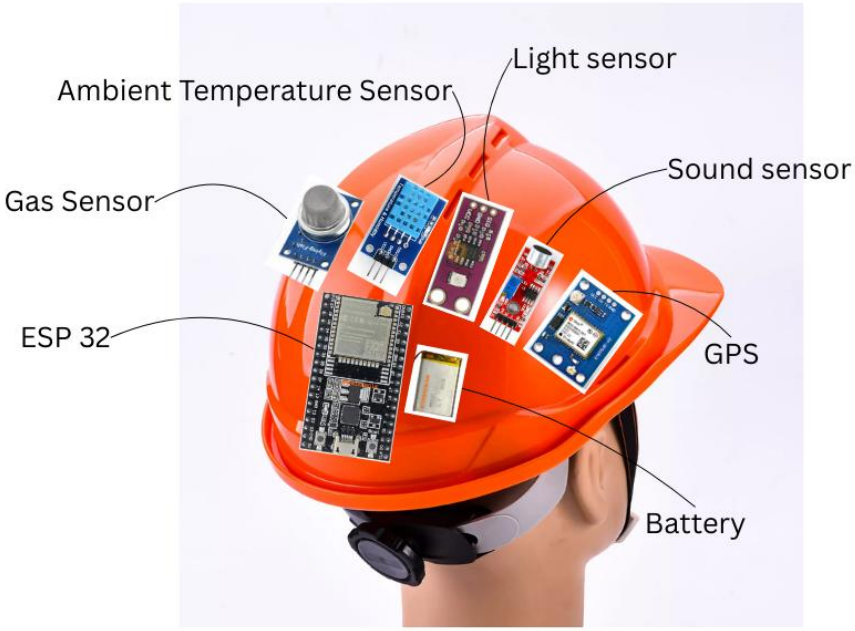


Figure 2: Helmet Design - Outer View

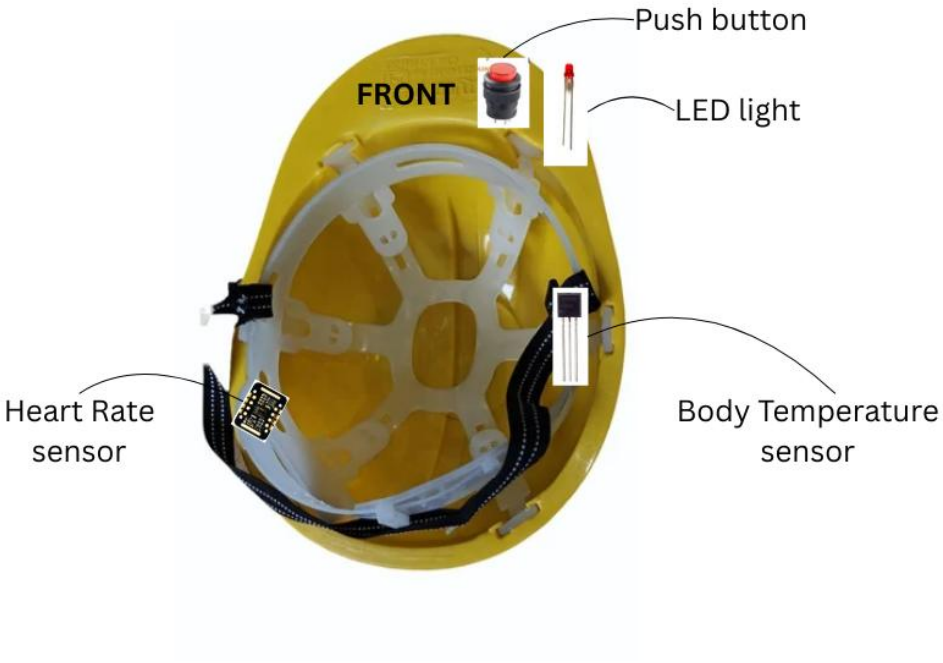


Figure 3: Helmet Design - Inner View